

Target validation paths — ulms-highgrade-pulm-mets-er-pr-neg-q qvt

The diagnostic and biomarker workup that hardens the targetable-feature call

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Target validation paths

Two test sets carry the weight of this case. An echocardiogram with LVEF and a clinician-documented ECOG decide whether the planned doxorubicin + trabectedin induction (the LMS-04 regimen, NCT02997358) can be given at all, and a tumor HRR/HRD panel decides whether olaparib + temozolomide (NCI protocol 10250, NCT03880019) exists as an option at progression. Neither result is on file. If the cardiac or pulmonary screen fails, the first-line fitness conversation changes and belongs to the treating team, not to Libby; if the HRR panel returns proficient, the olaparib + temozolomide branch closes for practical purposes. Nearly everything else below runs in parallel with induction and costs no treatment days.

uLMS histology

Three items are essential before cycle 1. A transthoracic echocardiogram with LVEF and strain (same day to 1 week), because the anthracycline cannot be dosed without a reference value. A clinician-documented ECOG, because the 1 on file is an intake assumption and her symptoms fit either 1 or 2, a difference that decides full-dose induction and most trial screens. And a hepatic-renal panel with creatine kinase, which sets the trabectedin starting dose and its dose-modification rules. The CBC with reticulocytes and ferritin needs an altitude-adjusted reference interval stated in the chart: at roughly 8,500 ft her baseline haemoglobin runs high, and a sea-level-normed count will under-read chemotherapy anaemia. Expert sarcoma pathology review of both specimens, with mitotic index, coagulative necrosis and FNCLCC grade, is the essential item that gates no drug: a 6.0 cm mass handled as a benign fibroid was very unlikely to be sampled at the one-block-per-centimetre standard, major revisions occur in about a quarter of sarcoma second opinions, and the deeper levels should be cut before the molecular orders exhaust the block.

Bilateral pulmonary pattern

PFTs with DLCO and oximetry recorded at residence altitude come before induction. Without them, an on-treatment desaturation cannot be split into progression, pneumonitis or altitude physiology, and the same numbers later price any lung-directed surgery or radiation. PD-L1 with CD3 and CD8 on the 2026 lung core answers whether the infiltrative, possibly inflammatory radiologic read reflects an immune infiltrate; it gates nothing on its own, and if the core runs scant, the MMR and B7-H3 stains take priority.

Unmeasured molecular landscape

One RNA-inclusive tumor NGS panel on the 2024 resection block does most of the work here. The HRR gene set with an LOH score gates olaparib + temozolomide via NCT03880019 (2-3 weeks, archival FFPE). RNA fusion calling must be specified on the requisition: NTRK breakpoints sit in large introns, a DNA-only panel returns a false negative that reads as a closed question, and a positive gates larotrectinib, entrectinib or repotrectinib under the tumor-agnostic approval. TMB from a validated comprehensive panel and the four-stain MMR IHC close the pembrolizumab doors (TMB ≥ 10 mut/Mb, or dMMR); both are rare in this histology and cheap to settle. B7-H3 IHC carries the best odds of a positive of any surface target (97% of soft-tissue sarcomas stained positive, 69% at high level) and gates screening for the B7-H3 ADC programs. The germline panel (BRCA1/2, PALB2, RAD51C/D, ATM, TP53 and the Lynch genes) goes out with the somatic order, after the consent conversation it actually is; ERN GENTURIS puts germline TP53 testing before genotoxic chemotherapy. HLA class I typing at allele-level resolution is one blood draw, with PRAME and MAGE-A4 IHC held as reflex stains until a restricting allele returns; that prices the TCR-T and ImmTAC class (IMA203, brenetafusp, the SURPASS programme) without spending

tissue speculatively. TP53, RB1, ATRX and MED12 ride along free on the panel, and paired ATRX IHC on both blocks answers whether the 2024 primary represents the 2026 disease for the price of two stains. One assay cannot be ordered at all: the RAD51 foci test, the functional readout that separated 11.2 from 5.4 months of PFS in the olaparib + temozolomide trial, has no CLIA provider anywhere. Do not hold the somatic panel for it; ask any PARP-inhibitor trial site whether they run it at screening.

ER/PR <1%

The <1% ER and PR result genuinely forecloses aromatase inhibition, since durable benefit with letrozole occurred only in tumours staining above 90%, but only if the stain was run on current disease with working controls. So the first step is a records one: which specimen was stained, which clone, and whether the internal control worked. A failed stain on the 2024 block reads identically to a true negative. If the result came off the 2024 primary, repeat ER and PR on the 2026 lung core with percentage and intensity reported; if it was already run on the lung core, skip the restain and leave the door closed.

The 2024-2026 interval

Two records requests, one letter to the 2024 institution. The September 2024 operative report and gross description settle whether the specimen was morcellated; morcellation of an unrecognised leiomyosarcoma disperses tumour through the peritoneum, so a yes means pelvic and peritoneal disease must be actively excluded rather than assumed absent. And the pre-treatment CT with named RECIST target lesions has to happen before cycle 1, because the clean baseline window closes the day induction starts. Any interval imaging on disc bounds the growth rate of the lung disease, the closest available measure of tempo in a therapy-naive patient.

Biomarkers measured, but not to decision resolution

This case has almost nothing in the category, and that is itself the finding. The biomarker survey handed eight questions across (HRR/HRD, NTRK fusion, MSI/MMR, TMB, HLA class I, B7-H3, PRAME and MAGE-A4), and every one arrived never measured or not assessable rather than measured at insufficient resolution; each has its closing assay in the sections above. The one result on file whose resolution is in question is the ER/PR <1% itself: the assay is decision-grade, the specimen provenance is not, and the records step above is what closes that gap.

Where to order these assays

The echocardiogram, PFTs and oximetry, ECOG documentation, staging CT and the two records requests are performed at or requested through the treating institution, so they carry no send-out provider row below.

Assay	Provider	Decision gated	Contact
Baseline comprehensive metabolic panel with bilirubin, alkaline phosphatase, transaminases, creatine kinase and calculated creatinine clearance	Labcorp Oncology (preferred)	Starting dose and hepatic dose-modification rules for trabectedin in the LMS-04 regimen.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167

Baseline comprehensive metabolic panel with bilirubin, alkaline phosphatase, transaminases, creatine kinase and calculated creatinine clearance	Quest Diagnostics	Starting dose and hepatic dose-modification rules for trabectedin in the LMS-04 regimen.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
Baseline CBC with differential, reticulocyte count and ferritin, interpreted against an altitude-adjusted haemoglobin reference	Labcorp Oncology (preferred)	Dose modification and transfusion thresholds during LMS-04 induction, corrected for chronic high-altitude erythrocytosis.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
Baseline CBC with differential, reticulocyte count and ferritin, interpreted against an altitude-adjusted haemoglobin reference	Quest Diagnostics	Dose modification and transfusion thresholds during LMS-04 induction, corrected for chronic high-altitude erythrocytosis.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
Tumor NGS covering BRCA1, BRCA2, PALB2, RAD51C, RAD51D, ATM and the wider HRR gene set, with a reported genomic-instability / LOH score	Caris Life Sciences (preferred) (MI Profile (whole exome plus whole transcriptome))	Olaparib plus temozolomide (NCI protocol 10250, NCT03880019) and PARP-inhibitor trial entry at progression.	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
Tumor NGS covering BRCA1, BRCA2, PALB2, RAD51C, RAD51D, ATM and the wider HRR gene set, with a reported genomic-instability / LOH score	Tempus AI (xT plus xR)	Olaparib plus temozolomide (NCI protocol 10250, NCT03880019) and PARP-inhibitor trial entry at progression.	test info · 600 West Chicago Avenue, Suite 510, Chicago, IL 60654 · 1-800-739-4137
Tumor NGS covering BRCA1, BRCA2, PALB2, RAD51C, RAD51D, ATM and the wider HRR gene set, with a reported genomic-instability / LOH score	Foundation Medicine (FoundationOne CDx)	Olaparib plus temozolomide (NCI protocol 10250, NCT03880019) and PARP-inhibitor trial entry at progression.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Tumor NGS covering BRCA1, BRCA2, PALB2, RAD51C, RAD51D, ATM and the wider HRR gene set, with a reported genomic-instability / LOH score	Myriad Genetics (MyChoice CDx)	Olaparib plus temozolomide (NCI protocol 10250, NCT03880019) and PARP-inhibitor trial entry at progression.	test info · 320 Wakara Way, Salt Lake City, UT 84108 · 1-800-469-7423
RNA-based fusion NGS panel covering NTRK1, NTRK2, NTRK3 (and RET, ALK, ROS1 in the same run)	Caris Life Sciences (preferred) (Whole transcriptome sequencing (MI Profile))	Larotrectinib, entrectinib or repotrectinib under the tumor-agnostic NTRK fusion indication.	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
RNA-based fusion NGS panel covering NTRK1, NTRK2, NTRK3 (and RET, ALK, ROS1 in the same run)	Tempus AI (xR RNA sequencing)	Larotrectinib, entrectinib or repotrectinib under the tumor-agnostic NTRK fusion indication.	test info · 600 West Chicago Avenue, Suite 510, Chicago, IL 60654 · 1-800-739-4137
RNA-based fusion NGS panel covering NTRK1, NTRK2, NTRK3 (and RET, ALK, ROS1 in the same run)	NeoGenomics Laboratories (NeoTYPE Fusion Profile (RNA))	Larotrectinib, entrectinib or repotrectinib under the tumor-agnostic NTRK fusion indication.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907

RNA-based fusion NGS panel covering NTRK1, NTRK2, NTRK3 (and RET, ALK, ROS1 in the same run)	Foundation Medicine (FoundationOne Heme)	Larotrectinib, entrectinib or repotrectinib under the tumor-agnostic NTRK fusion indication.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Mismatch-repair protein IHC (MLH1, PMS2, MSH2, MSH6), with reflex MLH1 promoter methylation and BRAF V600E if MLH1 is lost	NeoGenomics Laboratories (preferred) (MMR IHC panel)	Pembrolizumab or dostarlimab under the tumor-agnostic MSI-H / dMMR indication, and Lynch syndrome referral.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
Mismatch-repair protein IHC (MLH1, PMS2, MSH2, MSH6), with reflex MLH1 promoter methylation and BRAF V600E if MLH1 is lost	Labcorp Oncology	Pembrolizumab or dostarlimab under the tumor-agnostic MSI-H / dMMR indication, and Lynch syndrome referral.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
Mismatch-repair protein IHC (MLH1, PMS2, MSH2, MSH6), with reflex MLH1 promoter methylation and BRAF V600E if MLH1 is lost	Quest Diagnostics	Pembrolizumab or dostarlimab under the tumor-agnostic MSI-H / dMMR indication, and Lynch syndrome referral.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
Mismatch-repair protein IHC (MLH1, PMS2, MSH2, MSH6), with reflex MLH1 promoter methylation and BRAF V600E if MLH1 is lost	Mayo Clinic Laboratories (MMR IHC with reflex MLH1 methylation)	Pembrolizumab or dostarlimab under the tumor-agnostic MSI-H / dMMR indication, and Lynch syndrome referral.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
Mismatch-repair protein IHC (MLH1, PMS2, MSH2, MSH6), with reflex MLH1 promoter methylation and BRAF V600E if MLH1 is lost	ARUP Laboratories	Pembrolizumab or dostarlimab under the tumor-agnostic MSI-H / dMMR indication, and Lynch syndrome referral.	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 1-800-522-2787
Validated comprehensive genomic profiling panel reporting TMB in mutations per megabase (with MSI and the BRAF codon 600 call in the same report)	Foundation Medicine (preferred) (FoundationOne CDx)	Pembrolizumab under the tumor-agnostic TMB-H indication (KEYNOTE-158, NCT02628067).	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Validated comprehensive genomic profiling panel reporting TMB in mutations per megabase (with MSI and the BRAF codon 600 call in the same report)	Caris Life Sciences (MI Profile TMB)	Pembrolizumab under the tumor-agnostic TMB-H indication (KEYNOTE-158, NCT02628067).	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
Validated comprehensive genomic profiling panel reporting TMB in mutations per megabase (with MSI and the BRAF codon 600 call in the same report)	Tempus AI (xT)	Pembrolizumab under the tumor-agnostic TMB-H indication (KEYNOTE-158, NCT02628067).	test info · 600 West Chicago Avenue, Suite 510, Chicago, IL 60654 · 1-800-739-4137
Validated comprehensive genomic profiling panel reporting TMB in mutations per megabase (with MSI and the BRAF codon 600 call in the same report)	NeoGenomics Laboratories	Pembrolizumab under the tumor-agnostic TMB-H indication (KEYNOTE-158, NCT02628067).	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907

Expert sarcoma pathology review of both specimens with formal grading: mitotic index per 10 high-power fields, coagulative tumor cell necrosis, nuclear atypia, FNCLCC grade, plus an assessment of whether the 2024 block was adequately sampled	MD Anderson Cancer Center, Department of Pathology (preferred)	Formal grade and mitotic index for prognosis and for trial screening forms; also settles whether the 2024 block needs deeper levels.	test info · 1515 Holcombe Boulevard, Houston, TX 77030 · 1-877-632-6789
Expert sarcoma pathology review of both specimens with formal grading: mitotic index per 10 high-power fields, coagulative tumor cell necrosis, nuclear atypia, FNCLCC grade, plus an assessment of whether the 2024 block was adequately sampled	Memorial Sloan Kettering Cancer Center, Department of Pathology	Formal grade and mitotic index for prognosis and for trial screening forms; also settles whether the 2024 block needs deeper levels.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
Expert sarcoma pathology review of both specimens with formal grading: mitotic index per 10 high-power fields, coagulative tumor cell necrosis, nuclear atypia, FNCLCC grade, plus an assessment of whether the 2024 block was adequately sampled	Johns Hopkins Department of Pathology	Formal grade and mitotic index for prognosis and for trial screening forms; also settles whether the 2024 block needs deeper levels.	test info · 600 North Wolfe Street, Baltimore, MD 21287 · 1-410-955-5000
Expert sarcoma pathology review of both specimens with formal grading: mitotic index per 10 high-power fields, coagulative tumor cell necrosis, nuclear atypia, FNCLCC grade, plus an assessment of whether the 2024 block was adequately sampled	Mayo Clinic Laboratories (Pathology consultation service)	Formal grade and mitotic index for prognosis and for trial screening forms; also settles whether the 2024 block needs deeper levels.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
HLA class I genotyping at 4-digit (allele-level) resolution	Histogenetics (preferred) (Allele-level HLA class I typing)	PRAME, MAGE-A4 and NY-ESO-1 directed TCR-T and ImmTAC programs, all of which are allele-restricted.	test info · 300 Executive Boulevard, Ossining, NY 10562 · 1-914-762-0400
HLA class I genotyping at 4-digit (allele-level) resolution	Versiti Diagnostic Laboratories (HLA typing, high resolution)	PRAME, MAGE-A4 and NY-ESO-1 directed TCR-T and ImmTAC programs, all of which are allele-restricted.	test info · 638 North 18th Street, Milwaukee, WI 53233 · 1-800-245-3117
HLA class I genotyping at 4-digit (allele-level) resolution	Labcorp Oncology	PRAME, MAGE-A4 and NY-ESO-1 directed TCR-T and ImmTAC programs, all of which are allele-restricted.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
HLA class I genotyping at 4-digit (allele-level) resolution	ARUP Laboratories (HLA class I typing, high resolution)	PRAME, MAGE-A4 and NY-ESO-1 directed TCR-T and ImmTAC programs, all of which are allele-restricted.	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 1-800-522-2787

B7-H3 (CD276) IHC on tumor tissue	NeoGenomics Laboratories (preferred) (Custom IHC / pharma services)	Screening access to B7-H3-directed ADC, bispecific and CAR-T programs; trial-only, with no approved agent in this class.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
B7-H3 (CD276) IHC on tumor tissue	Discovery Life Sciences	Screening access to B7-H3-directed ADC, bispecific and CAR-T programs; trial-only, with no approved agent in this class.	test info · 601 Genome Way, Huntsville, AL 35806
B7-H3 (CD276) IHC on tumor tissue	MD Anderson Cancer Center, Department of Pathology	Screening access to B7-H3-directed ADC, bispecific and CAR-T programs; trial-only, with no approved agent in this class.	test info · 1515 Holcombe Boulevard, Houston, TX 77030 · 1-877-632-6789
Retrieval of the ER and PR IHC report: specimen of origin (2024 uterine resection versus 2026 lung core), antibody clone, scoring system and internal control adequacy	MD Anderson Cancer Center, Department of Pathology (preferred)	Whether the endocrine foreclosure holds, and therefore whether aromatase inhibition stays off the option list.	test info · 1515 Holcombe Boulevard, Houston, TX 77030 · 1-877-632-6789
TP53, RB1, ATRX and MED12 status from the comprehensive tumor NGS panel	Caris Life Sciences (preferred) (MI Profile)	Prognostic framing and biological classification; informs trial matching but gates no current agent.	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
TP53, RB1, ATRX and MED12 status from the comprehensive tumor NGS panel	Tempus AI (xT)	Prognostic framing and biological classification; informs trial matching but gates no current agent.	test info · 600 West Chicago Avenue, Suite 510, Chicago, IL 60654 · 1-800-739-4137
TP53, RB1, ATRX and MED12 status from the comprehensive tumor NGS panel	Foundation Medicine (FoundationOne CDx)	Prognostic framing and biological classification; informs trial matching but gates no current agent.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Germline multigene hereditary cancer panel including BRCA1, BRCA2, PALB2, RAD51C, RAD51D, ATM, TP53 and the Lynch genes	Myriad Genetics (preferred) (MyRisk Hereditary Cancer Test)	Cascade testing for relatives, Li-Fraumeni surveillance if TP53 is positive, and germline confirmation of any somatic HRR finding for PARP-inhibitor trial screening.	test info · 320 Wakara Way, Salt Lake City, UT 84108 · 1-800-469-7423
Germline multigene hereditary cancer panel including BRCA1, BRCA2, PALB2, RAD51C, RAD51D, ATM, TP53 and the Lynch genes	Ambry Genetics (CancerNext-Expanded)	Cascade testing for relatives, Li-Fraumeni surveillance if TP53 is positive, and germline confirmation of any somatic HRR finding for PARP-inhibitor trial screening.	test info · 1 Enterprise, Aliso Viejo, CA 92656 · 1-866-262-7943
Germline multigene hereditary cancer panel including BRCA1, BRCA2, PALB2, RAD51C, RAD51D, ATM, TP53 and the Lynch genes	GeneDx	Cascade testing for relatives, Li-Fraumeni surveillance if TP53 is positive, and germline confirmation of any somatic HRR finding for PARP-inhibitor trial screening.	test info · 207 Perry Parkway, Gaithersburg, MD 20877 · 1-888-729-1206

Germline multigene hereditary cancer panel including BRCA1, BRCA2, PALB2, RAD51C, RAD51D, ATM, TP53 and the Lynch genes	Labcorp Oncology (Labcorp hereditary cancer panel)	Cascade testing for relatives, Li-Fraumeni surveillance if TP53 is positive, and germline confirmation of any somatic HRR finding for PARP-inhibitor trial screening.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
Repeat ER and PR IHC on the 2026 pulmonary metastasis core, reported with percentage and intensity rather than a binary call	NeoGenomics Laboratories (preferred) (ER / PR IHC)	Whether aromatase inhibition re-enters the option list as a later-line or maintenance-phase choice.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
Repeat ER and PR IHC on the 2026 pulmonary metastasis core, reported with percentage and intensity rather than a binary call	Labcorp Oncology	Whether aromatase inhibition re-enters the option list as a later-line or maintenance-phase choice.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
Repeat ER and PR IHC on the 2026 pulmonary metastasis core, reported with percentage and intensity rather than a binary call	Quest Diagnostics	Whether aromatase inhibition re-enters the option list as a later-line or maintenance-phase choice.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
Repeat ER and PR IHC on the 2026 pulmonary metastasis core, reported with percentage and intensity rather than a binary call	Mayo Clinic Laboratories	Whether aromatase inhibition re-enters the option list as a later-line or maintenance-phase choice.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
PRAME IHC on archival tissue, ordered only if HLA class I typing returns a restricting allele	NeoGenomics Laboratories (preferred) (PRAME IHC)	PRAME-directed ImmTAC and TCR-T programs (brenetafusp, IMA203); trial-only, and conditional on HLA type.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
PRAME IHC on archival tissue, ordered only if HLA class I typing returns a restricting allele	Mayo Clinic Laboratories	PRAME-directed ImmTAC and TCR-T programs (brenetafusp, IMA203); trial-only, and conditional on HLA type.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
PRAME IHC on archival tissue, ordered only if HLA class I typing returns a restricting allele	Labcorp Oncology	PRAME-directed ImmTAC and TCR-T programs (brenetafusp, IMA203); trial-only, and conditional on HLA type.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
PRAME IHC on archival tissue, ordered only if HLA class I typing returns a restricting allele	Discovery Life Sciences	PRAME-directed ImmTAC and TCR-T programs (brenetafusp, IMA203); trial-only, and conditional on HLA type.	test info · 601 Genome Way, Huntsville, AL 35806
MAGE-A4 IHC on archival tissue, ordered only if HLA class I typing returns a restricting allele	NeoGenomics Laboratories (preferred) (MAGE-A4 IHC)	MAGE-A4-directed TCR-T trials (SURPASS programme); trial-only for this histology, and conditional on HLA type.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
MAGE-A4 IHC on archival tissue, ordered only if HLA class I typing returns a restricting allele	Mayo Clinic Laboratories	MAGE-A4-directed TCR-T trials (SURPASS programme); trial-only for this histology, and conditional on HLA type.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710

MAGE-A4 IHC on archival tissue, ordered only if HLA class I typing returns a restricting allele	Discovery Life Sciences	MAGE-A4-directed TCR-T trials (SURPASS programme); trial-only for this histology, and conditional on HLA type.	test info · 601 Genome Way, Huntsville, AL 35806
RAD51 foci immunofluorescence assay for functional homologous-recombination deficiency	Discovery Life Sciences (preferred)	Refines the olaparib plus temozolomide decision beyond HRR mutation status; supplements rather than replaces the somatic panel.	test info · 601 Genome Way, Huntsville, AL 35806
RAD51 foci immunofluorescence assay for functional homologous-recombination deficiency	MD Anderson Cancer Center, Department of Pathology	Refines the olaparib plus temozolomide decision beyond HRR mutation status; supplements rather than replaces the somatic panel.	test info · 1515 Holcombe Boulevard, Houston, TX 77030 · 1-877-632-6789
ATRX IHC on both the September 2024 primary and the 2026 pulmonary metastasis	NeoGenomics Laboratories (preferred) (ATRX IHC)	Whether the 2024 primary is representative of the 2026 metastatic disease; prognostic rather than treatment-selecting.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
ATRX IHC on both the September 2024 primary and the 2026 pulmonary metastasis	Mayo Clinic Laboratories	Whether the 2024 primary is representative of the 2026 metastatic disease; prognostic rather than treatment-selecting.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
ATRX IHC on both the September 2024 primary and the 2026 pulmonary metastasis	ARUP Laboratories	Whether the 2024 primary is representative of the 2026 metastatic disease; prognostic rather than treatment-selecting.	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 1-800-522-2787
ATRX IHC on both the September 2024 primary and the 2026 pulmonary metastasis	Labcorp Oncology	Whether the 2024 primary is representative of the 2026 metastatic disease; prognostic rather than treatment-selecting.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
PD-L1 IHC with the clone and scoring system matched to the intended agent, plus CD3 and CD8 T-cell infiltrate density, on the 2026 pulmonary metastasis core	NeoGenomics Laboratories (preferred) (PD-L1 IHC (22C3, 28-8, SP142, SP263) with CD3 / CD8)	Supporting information for checkpoint-blockade discussion and for interpreting the infiltrative radiologic appearance; gates nothing on its own.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
PD-L1 IHC with the clone and scoring system matched to the intended agent, plus CD3 and CD8 T-cell infiltrate density, on the 2026 pulmonary metastasis core	Labcorp Oncology	Supporting information for checkpoint-blockade discussion and for interpreting the infiltrative radiologic appearance; gates nothing on its own.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
PD-L1 IHC with the clone and scoring system matched to the intended agent, plus CD3 and CD8 T-cell infiltrate density, on the 2026 pulmonary metastasis core	Quest Diagnostics	Supporting information for checkpoint-blockade discussion and for interpreting the infiltrative radiologic appearance; gates nothing on its own.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378

PD-L1 IHC with the clone and scoring system matched to the intended agent, plus CD3 and CD8 T-cell infiltrate density, on the 2026 pulmonary metastasis core

Mayo Clinic Laboratories

Supporting information for checkpoint-blockade discussion and for interpreting the infiltrative radiologic appearance; gates nothing on its own.

test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710

Biomarker plan

Recommended assay	Rationale	Suggested assay provider	Tissue requirements
Baseline transthoracic echocardiogram with LVEF and global longitudinal strain	A documented pre-treatment LVEF is the precondition for giving doxorubicin, and the LMS-04 schedule carries the anthracycline to a cumulative dose that puts a patient into the cardiotoxicity surveillance band (PMID 35835135). ASCO and the 2022 ESC cardio-oncology guideline both call for baseline ventricular imaging, with strain where available, before anthracycline exposure (PMID 27918725; PMID 36017568). Nothing cardiac is on file here, so cycle 1 would otherwise be given with no reference value against which a later fall in function could be read.	Treating institution (echocardiography service; no send-out provider)	no tissue; transthoracic echocardiogram (5-10 mL blood only if natriuretic peptides are added)

Baseline comprehensive metabolic panel with bilirubin, alkaline phosphatase, transaminases, creatine kinase and calculated creatinine clearance	Trabectedin dosing and its hepatic dose-modification rules run off bilirubin, alkaline phosphatase and transaminases, and hepatotoxicity plus rhabdomyolysis were the dose-limiting toxicities in the registration trial (PMID 26371143). Doxorubicin is also dose-reduced for hyperbilirubinaemia. No laboratory value of any kind was supplied at intake, so there is currently no baseline against which a treatment-emergent transaminase rise could be interpreted.	Labcorp Oncology · test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167	5-10 mL whole blood (serum chemistry); no tumor tissue
Baseline CBC with differential, reticulocyte count and ferritin, interpreted against an altitude-adjusted haemoglobin reference	Residence at roughly 8,500 ft raises steady-state haemoglobin by an amount that varies by population and is not captured by sea-level reference ranges (PMID 31257609), so a sea-level-normed CBC will make chemotherapy-induced anaemia look milder than it is and will delay transfusion and dose-modification decisions. Grade 3-4 neutropenia and thrombocytopenia were both common on the doxorubicin plus trabectedin arm of LMS-04 (PMID 35835135), which makes the pre-treatment counts the reference for every subsequent dose decision. No blood count is on file.	Labcorp Oncology · test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167	3-5 mL whole blood (EDTA); no tumor tissue

Baseline pulmonary function tests with DLCO, plus resting and ambulatory pulse oximetry recorded at residence altitude	Bilateral infiltrative pulmonary disease in a never-smoker at 8,500 ft needs a measured baseline before induction, because a resting saturation that looks abnormal at sea-level norms may be her physiological steady state, and a saturation that looks acceptable may hide real loss of reserve. The same numbers later decide whether metastasectomy or SBRT is survivable once systemic control is achieved, which the consulting sarcoma oncologist explicitly kept on the table. Without a pre-treatment measurement, any on-treatment desaturation cannot be separated into progression, pneumonitis, or altitude physiology.	Treating institution (pulmonary function laboratory; record residence altitude)	no tissue; spirometry with DLCO and oximetry
Documented clinician-assessed ECOG performance status (with Karnofsky equivalent) recorded at the pre-treatment visit	The ECOG of 1 carried through this case was assigned at intake as an assumption, not recorded by a clinician, and the intake note says so explicitly. Nearly every sarcoma trial sets an ECOG ceiling of 1 or 2 at screening, so an undocumented score will either stall an application or produce a screen failure after tissue and time have been spent. It is free to fix and should be written into the chart at the same visit as the pre-induction labs.	Treating clinician, at the pre-treatment visit	no tissue; clinician assessment at the pre-treatment visit

Tumor NGS covering BRCA1, BRCA2, PALB2, RAD51C, RAD51D, ATM and the wider HRR gene set, with a reported genomic-instability / LOH score	Of everything untested here, status is the entry this histology most earns: olaparib plus temozolomide produced a 27% objective response rate and a median PFS of 6.9 months in pretreated advanced uterine leiomyosarcoma, and 31% of the enrolled patients carried an HRR gene mutation (PMID 37467452). Uterine leiomyosarcoma has a recognised HR-deficient subset including BRCA2 deletion and mutation (PMID 37143137). Skipping it leaves the one non-cytotoxic branch in this case unopened at the point where first-line chemotherapy fails.	Caris Life Sciences (MI Profile (whole exome plus whole transcriptome)) · test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669	archival FFPE acceptable; the 2024 resection block is preferred for tumor content and volume, with the 2026 lung core held in reserve
RNA-based fusion NGS panel covering NTRK1, NTRK2, NTRK3 (and RET, ALK, ROS1 in the same run)	This must be an RNA-based assay: NTRK breakpoints sit in large introns and a DNA-only panel misses them, so ordering comprehensive profiling without confirming RNA fusion calling would produce a false negative that reads as a closed question. Prevalence in uterine leiomyosarcoma is very low, but spindle-cell morphology is exactly the setting in which an NTRK-rearranged mesenchymal neoplasm is misclassified, and this tumor was already signed out as a benign leiomyoma once. A positive converts a chemotherapy-only plan into an oral TRK inhibitor with durable response rates.	Caris Life Sciences (Whole transcriptome sequencing (MI Profile)) · test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669	archival FFPE acceptable; runs on the same 2024 resection block as the DNA panel, no separate procedure

Mismatch-repair protein IHC (MLH1, PMS2, MSH2, MSH6), with reflex MLH1 promoter methylation and BRAF V600E if MLH1 is lost	Four stains on a block that is already being cut close a tumor-agnostic PD-1 indication that would otherwise sit open as a question at progression. Uterine leiomyosarcoma is nearly always mismatch-repair proficient, so a negative is the expected result and the value is in settling it cheaply and early. Loss of any one protein in a woman in her thirties also carries Lynch syndrome implications for her and for first-degree relatives, which is a second reason not to leave it unmeasured.	NeoGenomics Laboratories (MMR IHC panel) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	archival FFPE; 4-5 unstained slides from either block, bundled with the other IHC in one cut
Validated comprehensive genomic profiling panel reporting TMB in mutations per megabase (with MSI and the BRAF codon 600 call in the same report)	TMB here is unknown rather than low, because no sequencing of any kind has been performed. Leiomyosarcoma is copy-number driven and usually low-TMB, so a value at or above 10 mut/Mb would be unusual, but the same order returns MSI, BRAF V600E, the HRR genes and the TP53/RB1/ATRX/MED12 backbone, which makes TMB close to free once the panel is placed. Specify a validated comprehensive assay: a TMB figure from a small hotspot panel does not satisfy the tumor-agnostic indication and is not portable between platforms.	Foundation Medicine (FoundationOne CDx) · test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639	archival FFPE acceptable; same 2024 resection block and same accession as the HRR panel

Expert sarcoma pathology review of both specimens with formal grading: mitotic index per 10 high-power fields, coagulative tumor cell necrosis, nuclear atypia, FNCLCC grade, plus an assessment of whether the 2024 block was adequately sampled	The diagnosis itself has now been confirmed by a reference laboratory, but the parameters that drive prognosis and trial eligibility (mitotic count, coagulative necrosis, grade) are not on file, and a 6.0 cm myometrial mass handled as a benign leiomyoma in 2024 was very unlikely to have been sampled at the one block per centimetre standard applied to a suspected sarcoma. Major diagnostic revisions occur in a quarter of sarcoma cases sent for expert second opinion (PMID 33061792), and this case has already produced one. Without documented grade and mitotic index, several trial screening forms cannot be completed and the prognostic conversation rests on assumption.	MD Anderson Cancer Center, Department of Pathology · test info · 1515 Holcombe Boulevard, Houston, TX 77030 · 1-877-632-6789	archival FFPE blocks and slides from both specimens; additional levels may be cut from the 2024 block, which will consume tissue
HLA class I genotyping at 4-digit (allele-level) resolution	Every HLA-restricted agent that could reach this patient names its restricting allele, most often HLA-A*02:01, so without a genotype the PRAME and MAGE-A4 rows sit as unassessable rather than negative. It is a blood draw that spends no tumor tissue and costs little, and the result either opens the TCR-T and ImmTAC class for later consideration or closes it cleanly. Ordering it now avoids a two-step delay if she progresses through first-line chemotherapy while a trial slot is open.	Histogenetics (Allele-level HLA class I typing) · test info · 300 Executive Boulevard, Ossining, NY 10562 · 1-914-762-0400	5-10 mL whole blood (ACD or EDTA) or a buccal swab; no tumor tissue consumed

B7-H3 (CD276) IHC on tumor tissue	B7-H3 carries the best odds of a positive result of any surface target in this histology: 97% of soft tissue sarcomas stained positive and about 69% were high in a 153-patient series across 15 subtypes (PMID 39478506). Nothing is approved against it, so a positive buys trial access rather than a prescription, but the ADC and bispecific programs generally require documented expression at screening and a later referral should not stall while tissue is hunted down. Expression is spatially heterogeneous, which argues for reading the larger resection rather than the core alone.	NeoGenomics Laboratories (Custom IHC / pharma services) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	archival FFPE; 2-3 unstained slides, bundled into the same block cut as the MMR, PD-L1 and HER2 stains
Retrieval of the ER and PR IHC report: specimen of origin (2024 uterine resection versus 2026 lung core), antibody clone, scoring system and internal control adequacy	ER and PR are recorded as confirmed at <1%, and that negative closes a door which is genuinely open in this histology: letrozole met its activity criterion in ER and/or PR-positive uterine leiomyosarcoma, with the longest PFS in tumors staining strongly and diffusely (PMID 24222211). Before a foreclosure of that weight is relied on, the report should state which specimen was stained, which clone was used and whether the internal control worked, because a failed stain on an old block reads identically to a true negative at the <1% cutoff. This is a records step rather than a new assay, and it costs nothing but a phone call.	MD Anderson Cancer Center, Department of Pathology · test info · 1515 Holcombe Boulevard, Houston, TX 77030 · 1-877-632-6789	no new tissue; existing report and stained slides retrieved from the originating laboratory

TP53, RB1, ATRX and MED12 status from the comprehensive tumor NGS panel	These four genes are the mutational backbone of uterine leiomyosarcoma on exome sequencing (PMID 26891131) and they ride along free on the panel already being ordered for HRR, TMB and MSI. ATRX loss in particular tracks with the alternative lengthening of telomeres phenotype, aggressive histology and shorter survival in leiomyosarcoma (PMID 25229770), so it feeds the prognostic conversation and the urgency of the treatment plan. None of the four is directly druggable today, which is why this is a high rather than an essential row.	Caris Life Sciences (MI Profile) · test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669	no additional tissue; reported from the same 2024 resection accession as the HRR and TMB panel
Retrieval of the September 2024 operative report and gross pathology description: route of hysterectomy, whether power or manual morcellation was used, specimen integrity and margin status	A 6.0 cm myometrial mass removed as a presumed benign fibroid in 2024 may have been morcellated, and morcellation of an unrecognised leiomyosarcoma disperses tumor through the peritoneum and has been linked to worse outcomes (PMID 28209496; PMID 26646120). The current staging picture is pulmonary only; if the specimen was disrupted, peritoneal and pelvic disease has to be actively excluded rather than assumed absent. The answer sits in an operative note that nobody has yet had reason to re-read.	Records request to the 2024 institution (pair with the block release)	no tissue; operative note and gross description from the September 2024 hysterectomy

Pre-treatment CT chest, abdomen and pelvis with measurable RECIST target lesions, read against any imaging from the September 2024 to July 2026 interval	Induction has not started, so there is still a clean window to fix a pre-treatment baseline with named target lesions; once cycle 1 is given, that window is gone and response assessment becomes an argument about which prior scan counts. Any imaging from the 23-month interval would bound the growth rate of the pulmonary disease, which is the closest thing available to a measure of tempo in a systemic-therapy-naive patient. The infiltrative, possibly inflammatory appearance also needs to be documented before treatment, because it will complicate the reading of every subsequent scan.	Treating institution (radiology); request 2024-2026 outside imaging on disc	no tissue; contrast-enhanced CT, with prior outside imaging requested on disc
Germline multigene hereditary cancer panel including BRCA1, BRCA2, PALB2, RAD51C, RAD51D, ATM, TP53 and the Lynch genes	A germline result matters twice over here: it settles whether any somatic HRR finding is inherited, and at 39 with a high-grade sarcoma it carries direct implications for her siblings, children and any future pregnancy decisions. Uterine leiomyosarcoma is part of the Li-Fraumeni tumor spectrum, so TP53 belongs on the panel rather than being inferred from a somatic result. Somatic-only testing leaves this open, which is why the blood panel should go out alongside the tumor panel rather than as a reflex after it.	Myriad Genetics (MyRisk Hereditary Cancer Test) · test info · 320 Wakara Way, Salt Lake City, UT 84108 · 1-800-469-7423	5-10 mL whole blood (EDTA); drawn on the same visit as HLA typing

Repeat ER and PR IHC on the 2026 pulmonary metastasis core, reported with percentage and intensity rather than a binary call	If the <1% result came off the 2024 uterine primary, it describes a specimen taken 23 months and one metastatic step away from the disease being treated, and hormone receptor expression in uterine leiomyosarcoma is not uniformly retained between primary and metastasis. The letrozole trial selected on ER and/or PR positivity and found the longest PFS in strongly and diffusely positive tumors (PMID 24222211), so the question worth answering is not positive versus negative but whether any meaningful expression exists on current disease. If the original stain was already run on the lung core with working controls, skip this row.	NeoGenomics Laboratories (ER / PR IHC) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	archival FFPE; 2 unstained slides from the 2026 lung core, added to the same block cut as the other IHC
PRAME IHC on archival tissue, ordered only if HLA class I typing returns a restricting allele	PRAME cannot be called in either direction right now, because neither the antigen nor the restricting HLA allele has been tested. Sequence the two: if HLA typing returns A*02:01, PRAME IHC gates entry to the ImmTAC and TCR-T programs, and if it does not, this row closes with no tissue spent. Reported PRAME expression in leiomyosarcoma runs below the levels seen in synovial sarcoma, so a negative stain would not be surprising.	NeoGenomics Laboratories (PRAME IHC) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	archival FFPE; 2 unstained slides, ideally from the 2024 resection to preserve the lung core

MAGE-A4 IHC on archival tissue, ordered only if HLA class I typing returns a restricting allele	MAGE-A4 stays unknown until the HLA type is back, since the whole class is allele-restricted. Even a positive stain would not make her eligible for afamitresgene autoleucel, which was studied and licensed in synovial sarcoma and myxoid round cell liposarcoma rather than in sarcoma generally (PMID 38554725), so the realistic destination is a MAGE-A4 solid-tumor TCR-T trial of the kind run across histologies in HLA-A*02-positive patients (PMID 36624315). Stain it in the same antigen set as PRAME rather than as a separate order.	NeoGenomics Laboratories (MAGE-A4 IHC) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	archival FFPE; 2 unstained slides from the same cut as the PRAME stain
RAD51 foci assay for functional deficiency	In NCI protocol 10250 the RAD51 functional assay, not the mutation list, was what separated outcomes: median PFS was 11.2 months in tumors versus 5.4 months in proficient ones (PMID 37467452). That makes it the most direct read on whether this tumor would benefit from olaparib plus temozolomide, and it can be positive in tumors with no HRR gene mutation. Clinical availability is the problem: it is largely an academic and research assay with no widely offered CLIA version, which is why this ranks medium rather than essential.	Discovery Life Sciences · test info · 601 Genome Way, Huntsville, AL 35806	archival FFPE; feasibility should be confirmed with the receiving laboratory before slides are cut

ATRX IHC on both the September 2024 primary and the 2026 pulmonary metastasis	Paired sequencing of both blocks was floated and is not worth the cost here: no systemic therapy was given in the 23-month interval, so there is no treatment-driven selection pressure to have produced genomic divergence, and the 2024 resection has the tumor content the panel needs. Paired ATRX IHC is the cheap version of the same question, and ATRX loss with the ALT phenotype tracks with aggressive histology and shorter survival in leiomyosarcoma (PMID 25229770). Two stains on blocks already being cut will show whether the primary is representative of the metastatic disease.	NeoGenomics Laboratories (ATRX IHC) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	archival FFPE; 1-2 unstained slides from each block, bundled with the other IHC
PD-L1 IHC with the clone and scoring system matched to the intended agent, plus CD3 and CD8 T-cell infiltrate density, on the 2026 pulmonary metastasis core	PD-L1 gates no approval in soft tissue sarcoma, so this informs rather than decides, and the results that would actually open checkpoint blockade for her are MSI and TMB. It earns a row because the pulmonary lesions were read as infiltrative and possibly inflammatory, and someone will ask whether that appearance reflects an immune infiltrate or tumor growth pattern; a CD3 and CD8 read on the lung core answers that directly. Run it on the 2026 metastasis rather than the 2024 primary, since current immune phenotype is the point.	NeoGenomics Laboratories (PD-L1 IHC (22C3, 28-8, SP142, SP263) with CD3 / CD8) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	archival FFPE; 3-4 unstained slides from the 2026 lung core, bundled with the MMR and B7-H3 stains

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.